NU-SentEval at DialectSentEval Shared Task Subtask-1: Stacking Pre-trained Language Models with Focal Loss for Arabic Dialect Sentiment Analysis

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Abstract

This paper presents the system developed by the **NU-SentEval** team for Subtask 1 (Sentiment Analysis in Dialectal Arabic) of the DialectSentEval 2026 Shared Task. Arabic dialect sentiment classification is challenging due to morphological complexity, orthographic variation across regional dialects, and class imbalances in user reviews. To address these challenges, we propose an ensembled architecture leveraging five diverse Pre-trained Language Models (PLMs): MARBERTv2, MARBERT, CAMeLBERT-DA, AraBERTv0.2-Twitter, and Twitter-XLM-RoBERTa. All models were fine-tuned using a customized Focal Loss objective combined with Label Smoothing to mitigate majority class bias and prevent overconfidence. Out-of-fold probability representations from individual models were aggregated through a Logistic Regression stacking meta-classifier (`stacking_c1`). On the official test set, our system achieved a Macro F1-score of **0.8141** and an Accuracy of **0.8343**, demonstrating the effectiveness of combining specialized dialectal representations with loss regularization and meta-learning.

1 Introduction

Sentiment analysis of user-generated content in Arabic has become an essential area of research across social media monitoring, customer feedback analysis, and opinion mining. However, analyzing sentiment in Arabic is hindered by substantial linguistic heterogeneity. Modern Standard Arabic (MSA) differs considerably from the numerous regional dialects (e.g., Egyptian, Levantine, Gulf, and Maghrebi) commonly used across digital platforms. These dialects lack standardized orthography, feature non-conventional phonetic spellings, and frequently employ subtle cultural idioms, irony, and sarcasm.

To benchmark and advance modeling capabilities on dialectal Arabic text, the DialectSentEval 2026 Shared Task (Ezzini et al., 2026a) was organized at the ArabicNLP 2026 Conference. The shared task comprises two tracks: Subtask 1 (Arabic Dialect Sentiment Analysis) and Subtask 2 (Arabic Sentiment Swapping).

In this work, we describe our participation in **Subtask 1**. Our primary contribution is an end-to-end framework that couples comprehensive Arabic text normalization with an ensemble of five diverse Transformer-based language models. To tackle the challenge of class imbalance and boundary ambiguity, we train each model using Focal Loss regularized by Label Smoothing. Finally, we implement a Stacking meta-classifier that dynamically learns optimal decision boundaries across all model representations. Our final system delivered competitive performance on the official leaderboard, reaching a Macro F1-score of 0.8141.

2 Background

2.1 Task and Dataset Setup

The objective of Subtask 1 is multi-class sentiment classification for dialectal Arabic. The provided training dataset consists of 1,731 hotel and accommodation reviews written in four major Arabic dialect groups: Egyptian (432 samples), Moroccan Darija (433 samples), Saudi/Gulf (433 samples), and Jordanian/Levantine (433 samples). The dataset is annotated across three sentiment classes: Negative (657 samples, 38.0%), Positive (609 samples, 35.2%), and Neutral (465 samples, 26.9%).

Given an input review x, the system must predict a discrete sentiment label $y \in \{\text{Positive}, \text{Negative}, \text{Neutral}\}$. Representative examples from the dataset include:

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- **Input (Positive):** الفطور ممتاز، والخدمة رائعة والطاقم كايعاون (*"The breakfast is excellent, the service is wonderful, and the staff is helpful"*) → **Output:** Positive

- **Input (Negative):** ماكاينش إنترنت ف الغرفة ، الواي فاي غالي بزاف (*"There is no internet in the room, Wi-Fi is very expensive"*) → **Output:** Negative

- **Input (Neutral):** الوصول والخروج كانو مزيانين (*"Check-in and check-out were fine"*) → **Output:** Neutral

2.2 Related Work

Recent years have witnessed concerted efforts toward benchmarking sentiment analysis on dialectal Arabic. The AraSentEval shared task (Ezzini et al., 2026b) evaluated sentiment classification and dialectal swapping across regional varieties. Similarly, the ADOR benchmark (Alharbi et al., 2025c) provided a human-annotated corpus of multi-dialectal Arabic hotel reviews, highlighting the performance drop of general-purpose models when transferred to informal regional speech. The Ahasis shared task (Alharbi et al., 2025b) further demonstrated the utility of pre-trained transformer backbones fine-tuned on targeted dialectal benchmarks.

More recently, Alharbi et al. (2025a) evaluated Large Language Models (LLMs) on dialectal Arabic sentiment analysis, showing that specialized Arabic encoder models fine-tuned with domain-adapted objectives frequently match or surpass general closed-source LLMs while maintaining significantly higher inference efficiency. Building upon these insights, our work focuses on maximizing the collective representational strength of multiple dialect-adapted transformer models.

3 System Overview

Our proposed pipeline consists of three core components: (1) Domain-specific Text Normalization, (2) Diverse Base Model Fine-Tuning with Focal Loss, and (3) Logistic Regression Stacking Meta-Classification.

3.1 Text Preprocessing

Social media reviews often contain non-standard punctuation, decorative elongations, and platform-specific entities. We apply a targeted normalization pipeline:

1. **Diacritic & Tatweel Removal:** Stripping short vowel marks (Tashkeel) and character stretchers (Tatweel: `\u0640`).

2. **Letter Normalization:** Unifying all forms of Alef with Hamza to bare Alef (ا), replacing Alif Maksura (ى) with Yaa (ي), and Taa Marbuta (ة) with Haa (ه).

3. **Entity Masking:** Replacing user handles (`@username`) with `@USER` and hyperlinks with `URL`.

4. **Noise Reduction:** Collapsing repeating characters to a maximum of two consecutive letters and removing standalone punctuation.

3.2 Base Language Models

To ensure complementary linguistic representations across multiple Arabic dialects, we select five Pre-trained Language Models (PLMs):

1. **MARBERTv2** and **MARBERT**: Masked language models pre-trained on massive collections of Arabic tweets, providing strong dialectal coverage. in our evaluation, Marbert achieved macro F1 of 0.9461, and marbert v2 of macro f1 score of 0.9264.

2. **CAMeLBERT-DA**: A dedicated BERT variant pre-trained on Dialectal Arabic texts. In our evaluation,CAMeLBERT-DA achieved macro F1 of 0.9249.

3. **AraBERTv0.2-Twitter**: Pre-trained on Arabic news and millions of social media tweets,

and in our evaluation it achieved macro F1 score of 0.9599 which was the highest score among the other models.

4. **Twitter-XLM-RoBERTa**: A cross-lingual RoBERTa model adapted on multilingual Twitter streams.This model achieved macro F1 score of 0.9198.

3.3 Focal Loss with Label Smoothing

To address class imbalance and reduce overconfident probabilities on ambiguous samples, we implement a customized Focal Loss with Label Smoothing. Given predicted class probabilities $p \in [0, 1]^K$ and ground truth class y, standard cross-entropy is replaced by:

$$\mathcal{L}_{\text{FL}} = -\sum_{k=1}^{K} \tilde{y}_k (1 - p_k)^{\gamma} \log(p_k) \qquad (1)$$

where $\gamma = 2.0$ is the focusing parameter that down-weights easy examples, and $\tilde{y}_k$ represents the smoothed target distribution:

$$\tilde{y}_k = (1 - \epsilon) \cdot I(y = k) + \frac{\epsilon}{K} \qquad (2)$$

with label smoothing factor $\epsilon = 0.1$ and $K = 3$ sentiment classes.

3.4 Stacking Meta-Classifier (`stacking_c1`)

Instead of heuristic voting, we construct a second-level Meta-Classifier. For each sample x, the predicted logit vectors from all $M = 5$ base models are concatenated into a joint meta-feature vector $z = [z^{(1)}, z^{(2)}, \ldots, z^{(5)}] \in R^{15}$. We train an L_2-regularized Logistic Regression model ($C = 1.0$) on the validation logit representations to learn optimal model confidence weights and decision boundaries.

4 Experimental Setup

4.1 Training Strategy and Data Splits

We structured our training pipeline into two sequential phases:

- **Phase 1 (Validation & Epoch Tuning):** The dataset was partitioned into an 85% training split and a 15% stratified validation split. Each model was trained for up to 15 epochs with early stopping (patience = 3 based on validation Macro F1). The validation predictions were harvested to train the stacking meta-classifier.

Metric	Official Test Score
Macro F1-Score	**0.8141**
Accuracy	0.8343
Macro Precision	0.8115
Macro Recall	0.8197

Table 1: Official evaluation results of the NU-SentEval system on the Subtask 1 test set.

- **Phase 2 (Full Re-training):** With the optimal epoch count established for each base model, all five models were freshly re-initialized and re-trained on 100% of the provided training data before generating final test set predictions.

4.2 Hyperparameters and Environment

All models were implemented using PyTorch 2.x and Hugging Face `transformers` alongside `scikit-learn`. Models were fine-tuned with a maximum token length of 256, batch size of 16 with gradient accumulation steps of 2 (effective batch size = 32), AdamW optimizer with initial learning rate $\eta = 2 \times 10^{-5}$, 150 linear warmup steps, cosine learning rate scheduling, and weight decay of 0.01.

4.3 Evaluation Metrics

Following official task guidelines, performance is evaluated using **Macro F1-Score** as the primary metric, alongside Accuracy, Macro Precision, and Macro Recall.

5 Results

5.1 Official Test Results

Table 1 displays the official performance of our system on the unlabelled test set of DialectSentEval 2026 Subtask 1. Our ensemble system achieved a competitive Macro F1-score of **0.8141** and an Accuracy of **0.8343**.

5.2 Ablation Study of Ensemble Strategies

To quantify the benefit of our stacking meta-classifier, we evaluated several ensembling schemes on our Phase 1 validation split. As detailed in Table 2, the Logistic Regression meta-classifier (`stacking_c1`) achieved the highest performance across all evaluation metrics, outperforming top-k weighted averaging, power weighting, and temperature-scaled heuristics.

Strategy	Acc.	Prec.	Rec.	Macro F1
stacking_c1	**0.9654**	**0.9644**	**0.9621**	**0.9632**
top3_weighted	0.9615	0.9597	0.9585	0.9591
power_weighted	0.9577	0.9561	0.9551	0.9556
temp_f1_weighted	0.9538	0.9524	0.9503	0.9513

Table 2: Comparison of ensembling strategies on the Phase 1 validation split.

5.3 Class-wise Performance and Error Analysis

Table 3 provides the detailed class-level breakdown for the best-performing stacking_c1 model on the validation split.

Class	Prec.	Rec.	F1-Score	Support
Positive (0)	0.9700	0.9798	0.9749	99
Negative (1)	0.9559	0.9286	0.9420	70
Neutral (2)	0.9674	0.9780	0.9727	91
Macro Avg	**0.9644**	**0.9621**	**0.9632**	260

Table 3: Class-wise classification report of stacking_c1 on the Phase 1 validation split.

A closer inspection of model errors revealed two prominent failure patterns:

1. **Sarcasm and Polite Dissatisfaction:** In several customer reviews, users employed positive blessing phrases (e.g., الله يعطيكم العافية بس الخدمة ضايعة, meaning *"May God give you health, but the service is lost"*) to deliver negative feedback. In borderline cases, the strong positive sentiment lexicons biased base models toward Positive or Neutral.

2. **Mixed Dialectal Colloquialisms:** Infrequent regional slang words (e.g., specialized Maghrebi terms like ماشي مزيان or بزاف) occasionally led to dispersed probability distributions among models. The stacking meta-classifier successfully resolved the majority of these ambiguities by leaning on the most confident dialect-specific backbones (such as CAMeLBERT-DA and MARBERTv2).

6 Conclusion

In this paper, we detailed the NU-SentEval system for Subtask 1 of the DialectSentEval 2026 Shared Task. By combining five specialized Arabic Pre-trained Language Models with Focal Loss, Label Smoothing, and a Logistic Regression stacking meta-classifier, our approach demonstrated robust performance on diverse dialectal sentiment inputs, securing a test Macro F1-score of 0.8141. Future work will investigate dialect identification pre-filtering and synthetic data augmentation using generative LLMs.

Limitations

While our ensemble framework provides strong empirical accuracy, ensembling five large Transformer architectures increases inference latency and memory requirements compared to a single distilled model. Additionally, performance on highly localized colloquialisms remains dependent on pre-training corpus representation.

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